

# FIN5545 Project Part B: Systematic Multi-Asset Funds with News Sentiment

This report draft documents a reproducible walk-forward experiment over the 2020-2023 course dataset. The investment universe combines 50 US equities and 10 cryptocurrencies, with equity-only and crypto-only funds included for benchmarking. All portfolio weights are estimated using only trailing observations available at each monthly rebalance.

## 1. Fund design and backtest controls

The methods are equal weight, minimum variance, maximum Sharpe and inverse-volatility risk parity. Long-only weights are capped at 20 percent to reduce single-name concentration. Combined Equal Weight: first live date 2021-01-04, lookback 252 days, annualisation 252; Combined Max Sharpe: first live date 2021-01-04, lookback 252 days, annualisation 252; Combined Min Variance: first live date 2021-01-04, lookback 252 days, annualisation 252; Crypto Equal Weight: first live date 2021-01-01, lookback 365 days, annualisation 365; Crypto Risk Parity: first live date 2021-01-01, lookback 365 days, annualisation 365; Equity Equal Weight: first live date 2021-01-04, lookback 252 days, annualisation 252; Equity Max Sharpe: first live date 2021-01-04, lookback 252 days, annualisation 252; Equity Min Variance: first live date 2021-01-04, lookback 252 days, annualisation 252; Equity Sentiment Tilt: first live date 2021-01-04, lookback 252 days, annualisation 252.

Crypto returns are computed on the 365-day crypto calendar first. Combined funds then left-merge those already-computed crypto returns onto the equity trading calendar, matching the course calendar instruction.

## 2. Out-of-sample results

| Fund                  | Ann. return | Ann. vol | Sharpe | Max DD  |
|-----------------------|-------------|----------|--------|---------|
| Combined Equal Weight | 14.98%      | 21.25%   | 0.70   | -28.75% |
| Combined Max Sharpe   | 18.79%      | 18.83%   | 1.00   | -24.53% |
| Combined Min Variance | 6.83%       | 13.23%   | 0.52   | -17.63% |
| Crypto Equal Weight   | 34.19%      | 80.42%   | 0.43   | -81.60% |
| Crypto Risk Parity    | 38.56%      | 78.66%   | 0.49   | -80.21% |
| Equity Equal Weight   | 12.64%      | 16.17%   | 0.78   | -20.32% |
| Equity Max Sharpe     | 9.43%       | 15.71%   | 0.60   | -23.60% |
| Equity Min Variance   | 5.64%       | 12.93%   | 0.44   | -16.98% |
| Equity Sentiment Tilt | 9.22%       | 16.14%   | 0.57   | -20.27% |

The highest Sharpe in this run is Combined Max Sharpe at 1.00. This should be interpreted as an out-of-sample historical result, not a forecast. Differences across funds largely reflect the trade-off between volatility control and exposure to high-return crypto assets.

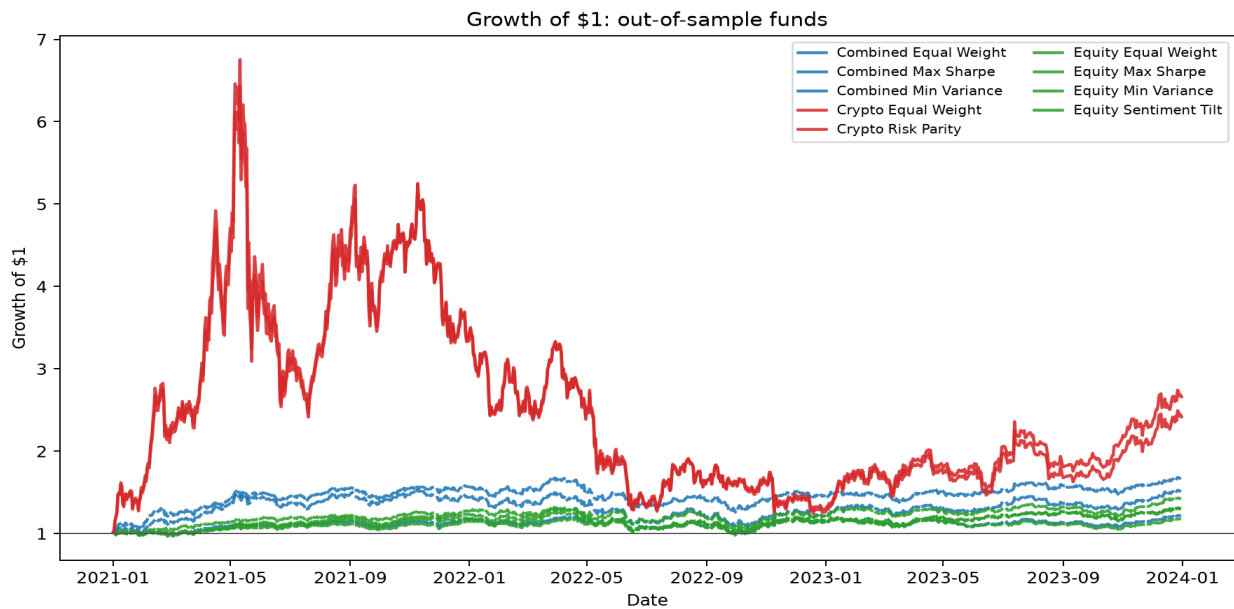


Figure 1. Growth of \$1 across all out-of-sample funds.

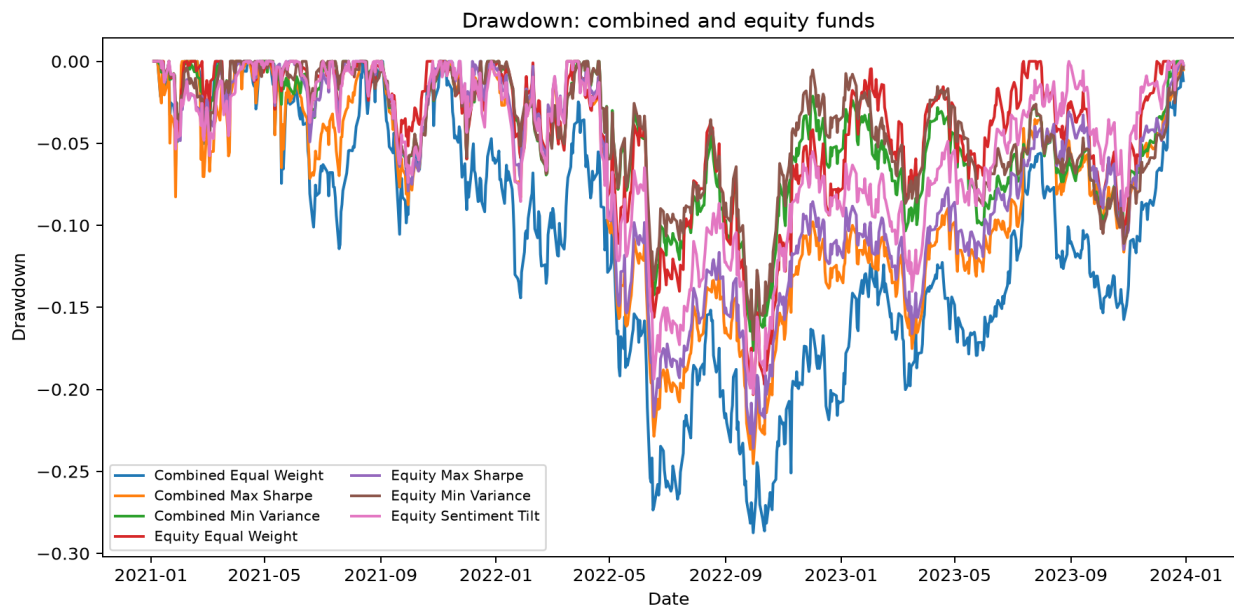


Figure 2. Drawdown paths for the main combined and equity funds.

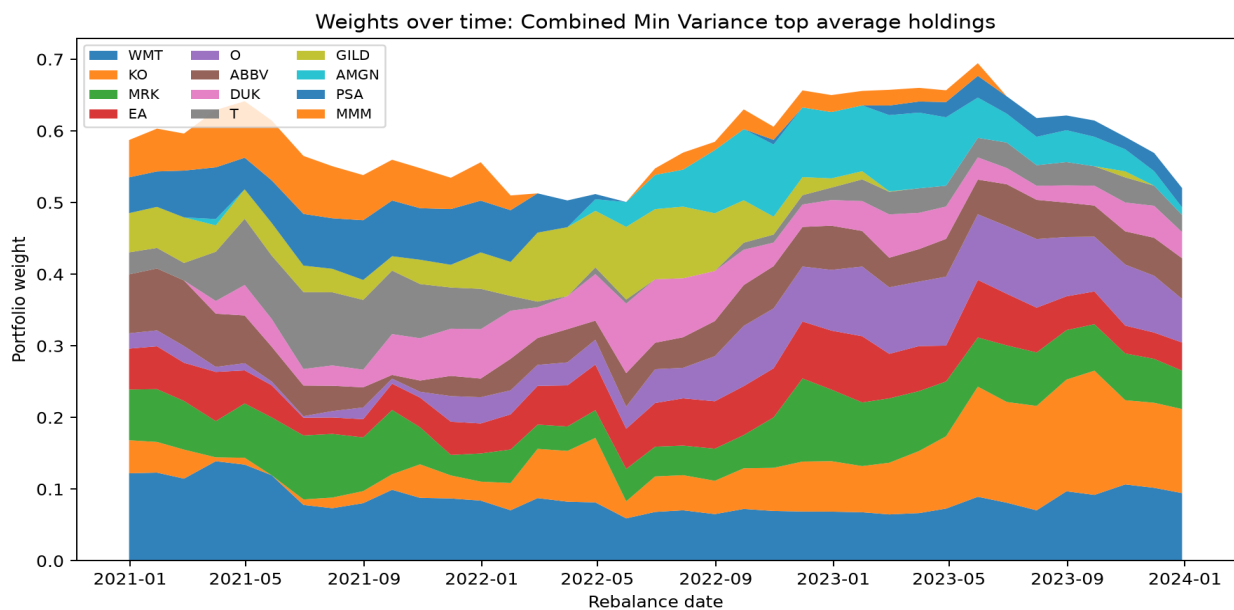


Figure 3. Top average holdings through time for the combined minimum-variance fund.

### 3. News sentiment index

Headlines are de-duplicated by ticker, date and title, then aligned to the same or next equity trading day. A transparent finance lexicon scores the assembled ticker-day text. The sector index is equal-weighted across stocks, smoothed over five trading days, and lagged by one trading day before use in any portfolio rule.

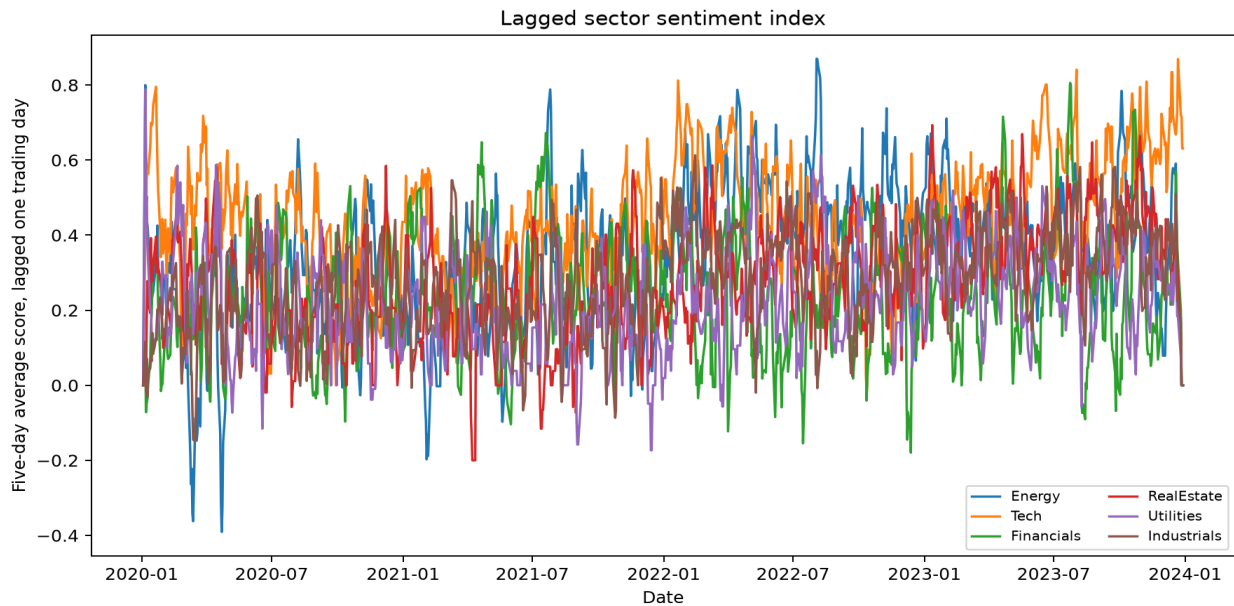


Figure 4. Lagged sector sentiment index for the six most variable sectors.

### 4. Fusion extension

The fusion test starts from the equity maximum-Sharpe weight schedule and tilts sector weights toward sectors with stronger lagged sentiment. Because the signal is lagged and measured before portfolio returns are realised, the test is look-ahead safe.

| Fund                  | Ann. return | Ann. vol | Sharpe | Max DD  |
|-----------------------|-------------|----------|--------|---------|
| Equity Max Sharpe     | 9.43%       | 15.71%   | 0.60   | -23.60% |
| Equity Sentiment Tilt | 9.22%       | 16.14%   | 0.57   | -20.27% |

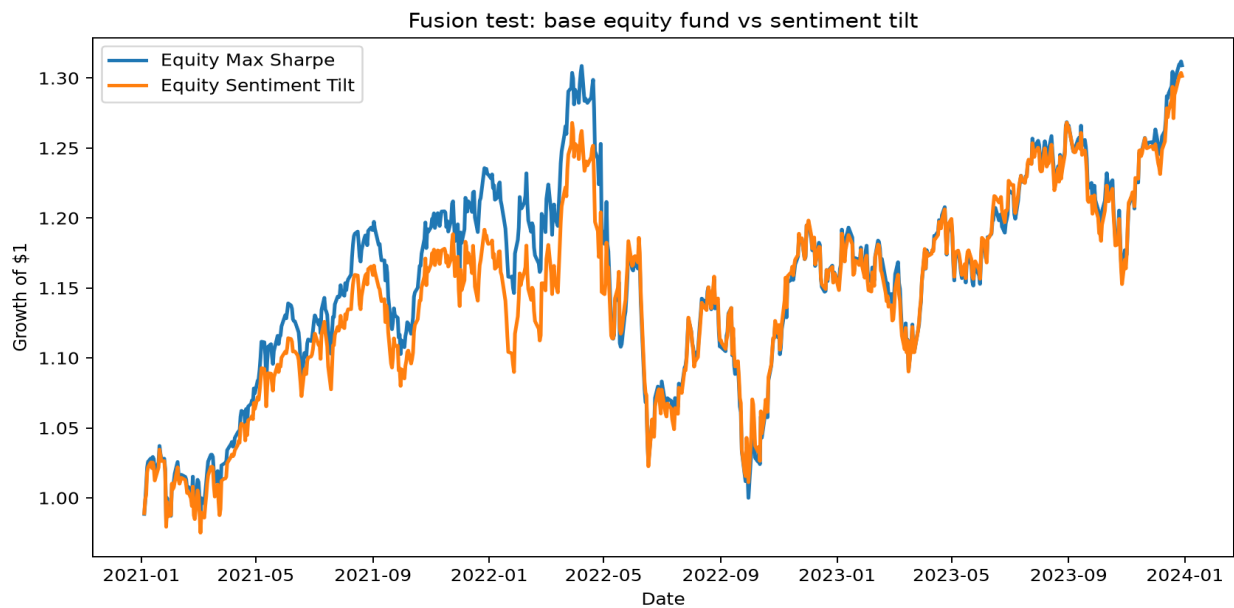


Figure 5. Before-vs-after comparison for the sentiment fusion rule.

### 5. App and recommendations

The Streamlit app reads only precomputed CSV files from results/. It supports fund comparison, fund fact sheets, a simple allocation blend, current holdings and sector sentiment analytics. This design keeps the deployed app lightweight and avoids running NLP or portfolio optimisation on Streamlit Community Cloud.

Recommendations: first, review the cap and rebalance frequency as sensitivity checks; second, treat the lexicon sentiment as a noisy proxy and compare it with VADER or a transformer model if time permits; third, report both positive and negative fusion results honestly because a weak signal is still informative for allocation governance.